CellSense AI

Industrial EV Fleet Asset Intelligence

ET AI Hackathon 2026 (2.0) · Detailed Solution Document

Problem Statement 3 — AI for Industrial EV Supply Chain & Asset Intelligence: Accelerating Net Zero

Theme: Sustainability / EV Manufacturing / Asset Performance Management / Supply Chain

Contents

1. Executive Summary

CellSense AI is an Asset Performance Management (APM) platform for industrial and commercial electric-vehicle fleets. It predicts each battery asset's State of Health (SoH) and Remaining Useful Life (RUL) from real cell-cycling data, ranks the fleet by risk, and — critically — converts every prediction into a grounded, cited maintenance decision using a multi-agent reasoning layer over OEM manuals and safety standards.

The project deliberately targets the operational bottleneck identified in Problem Statement 3: industrial fleet operators lack the asset-intelligence tooling to manage EV batteries with the same rigour they apply to conventional equipment. CellSense AI treats the battery as the asset, and applies predictive maintenance plus decision intelligence to it.

Our design principle is feasibility-first: a provable machine-learning core (measured against observed degradation) fused with a natural-language decision layer that no generic battery model provides. This combination maximises the two heaviest judging criteria — Innovation and Business Impact — while remaining fully buildable within the hackathon timeline on free, public data.

At a glance	
What it does	Predicts battery SoH & RUL, ranks fleet risk, generates cited maintenance actions
Core innovation	Decision-Intelligence layer — prediction becomes a grounded, explainable action
Provable metric	RUL / SoH prediction error (RMSE) vs. observed degradation
Data	Real NASA PCoE battery cells (34 cells; core B0005/06/07/18) — 100% real, no synthetic data in the model
Cost to build	Zero — free datasets, open-source stack, free-tier or local LLM

2. Problem Understanding

2.1 Context

India registered over 2 million EVs in FY2025, yet penetration in industrial and commercial segments — freight, mining, intra-plant logistics, construction equipment — remains below 2.5%. The barrier is no longer purely financial: incentives and total-cost-of-ownership parity with diesel are approaching. The real bottleneck is operational — operators cannot manage battery lifecycle, degradation, and maintenance with the discipline they apply to conventional industrial assets. Meeting India's 30% commercial-EV penetration target by 2030 depends on closing this asset-intelligence gap.

2.2 Core Problem

The battery is the single most valuable and most failure-prone asset in an industrial EV, yet its degradation is invisible until it fails. Operators lack (a) reliable remaining-life visibility, (b) fleet-level prioritisation of which asset to act on first, and (c) actionable, trustworthy guidance on what to do about it. The result is unplanned downtime and premature, capital-intensive battery replacement.

2.3 Root Causes

- **Modelling gap —** Battery degradation is non-linear and difficult to infer from raw telemetry without modelling.
- **Decision gap —** A predicted number alone is not a decision; operators need cause, severity, and a recommended action.

- **Knowledge gap —** OEM procedures, failure modes, warranty terms, and safety standards are fragmented and un-actioned at the point of need.

2.4 Pain Points

Pain point	Consequence
No remaining-useful-life visibility	Failures happen without warning; no planning horizon
Premature battery replacement	Wasted capital — the battery is 30–40% of vehicle cost
No fleet-level prioritisation	Maintenance effort is spread thin, not risk-targeted
Fragmented maintenance knowledge	Slow, inconsistent decisions; safety exposure
Thermal / anomaly events missed	Safety incidents and unplanned outages

2.5 Stakeholders

Stakeholder	What they gain
Fleet operations manager	Fleet-wide risk visibility and prioritised action list
Maintenance engineer	Predicted failures + grounded, cited repair/replace guidance
Sustainability / net-zero officer	Asset-level carbon and electrification progress tracking
CFO / capital planning	Avoided premature replacement; predictable capital cycles
OEM / battery supplier	Field degradation intelligence and warranty insight

3. Proposed Solution — CellSense AI

3.1 Overview

CellSense AI ingests battery-cycling data for each fleet asset, predicts its current SoH and forecasts its RUL, applies a deterministic rule engine to band risk (Healthy / Watch / Critical), and presents a risk-ranked fleet dashboard. When an operator opens a flagged asset, a three-agent pipeline diagnoses the probable cause, retrieves the relevant procedures from a document corpus, and produces a cited maintenance recommendation with an exportable report.

3.2 Core Innovation — Decision Intelligence over Prediction

Most battery projects stop at a prediction. CellSense AI closes the loop from prediction to grounded decision. The machine-learning core produces the number; a Retrieval-Augmented reasoning layer turns that number into an APM decision — for example: "Asset #14 crosses its degradation threshold in ~35 cycles; per maintenance procedure §4.2, schedule cell replacement before the next high-load shift." This is exactly the Asset Performance Management value proposition, delivered with citations for trust and auditability.

3.3 Key Differentiators

- **Trustworthy core —** Provable ML on real cell data — credibility, not a mock dashboard.
- **Explainable & safe —** RAG-grounded, cited recommendations — no hallucinated safety advice.
- **Decision-first —** Fleet-level prioritisation — ranks assets by degradation and operational criticality.
- **Domain-aligned —** Speaks the language of industrial Asset Performance Management end-to-end.

4. Scope & Feasibility Strategy

Problem Statement 3 is intentionally broad, spanning two prongs — fleet asset management and manufacturing supply chain. Attempting both fragments the narrative and the timeline. We commit fully to the fleet-APM prong and cut the rest deliberately.

Priority	Component	Decision
Must Have	Battery SoH + RUL prediction (ML core)	Build — provable metric
Must Have	Fleet risk-ranking dashboard	Build — decision surface
Must Have	RAG-grounded maintenance recommendation (agents)	Build — the innovation
Good to Have	Fleet electrification-readiness score + carbon stat	Only if core is demo-solid
Cut	Manufacturing quality intelligence (QMS/SPC)	Different domain — scatter
Cut	Multi-tier material supply-chain risk (Li/Co)	Separate data pipeline — scatter
Cut	Charging-infrastructure optimisation	Low demo payoff
Cut	Full digital twin / knowledge graph	Over-engineering — no marginal score

Rationale: one sharp, complete story scores higher than two blurry, half-finished ones. Every cut component would consume a full workstream while adding no marginal judging points beyond what the fleet-APM story already earns.

5. System Architecture

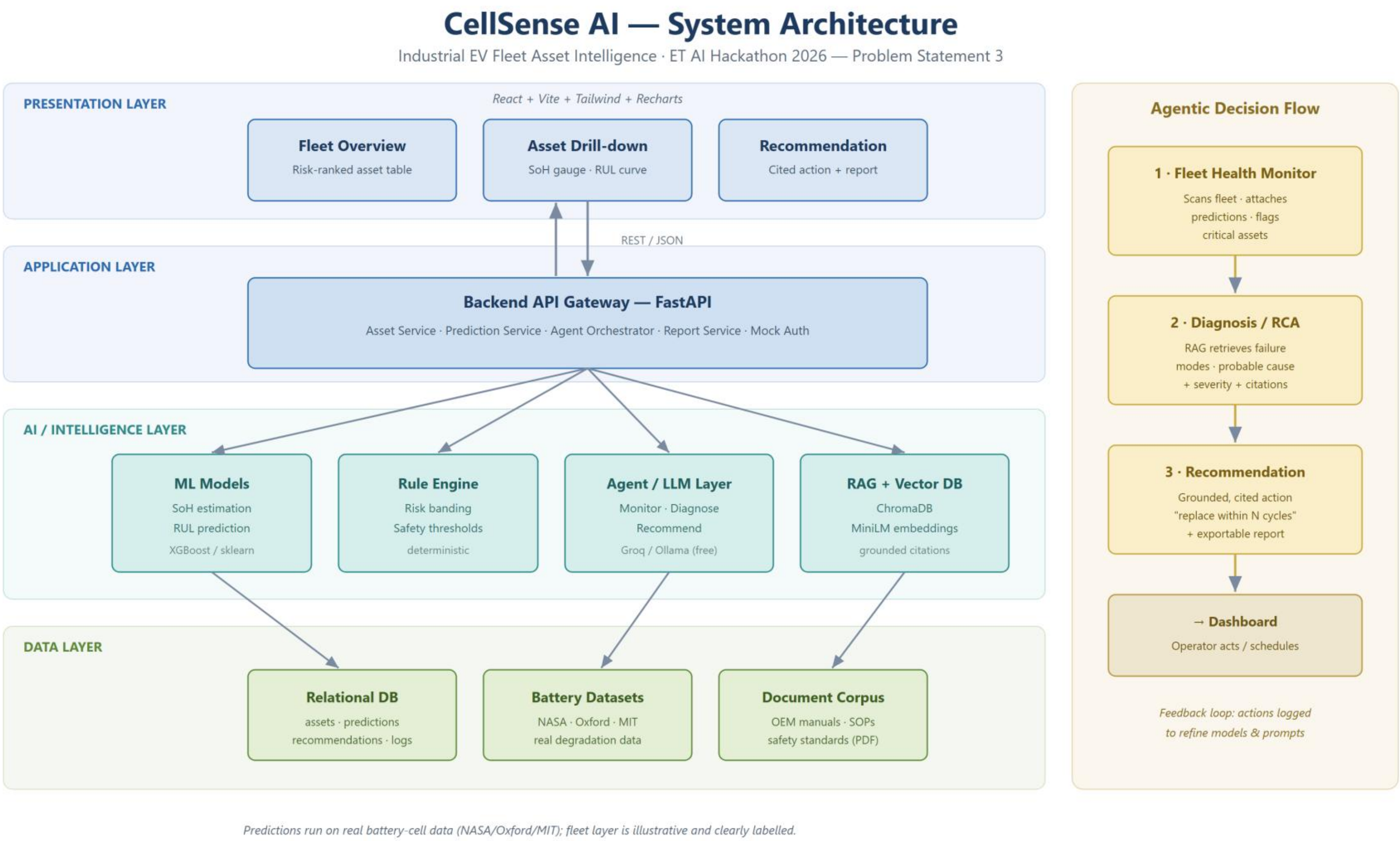

Figure 1 — CellSense AI layered system architecture and agentic decision flow.

5.1 Component Responsibilities

Layer / Component	Responsibility
Frontend (React)	Renders fleet table, per-asset SoH/RUL charts, recommendation panel. No business logic.
Backend (FastAPI)	Single API gateway; loads models, serves predictions, runs rule engine, orchestrates agents, persists results.
ML Models	Offline-trained SoH & RUL models loaded as artifacts; batch inference across the fleet.
Rule Engine	Deterministic risk banding and safety thresholds — owns all safety-critical decisions (no LLM).
Agent / LLM Layer	Three agents reason over predictions + retrieved documents to produce cited recommendations.
RAG + Vector DB	Embeds and retrieves OEM/procedure chunks (ChromaDB) to ground every recommendation.
Relational DB	System of record: assets, telemetry, predictions, recommendations, feedback log.
Auth	Mock single-role session only — deliberately out of scope for the prototype.

6. End-to-End Pipeline

The system runs in two modes: batch inference across the whole fleet, and on-demand agentic reasoning when an operator opens a flagged asset (keeping LLM calls minimal and the demo fast).

Battery cycle data + fleet metadata (Input) ↓

Data Ingestion — load datasets, normalise schema ↓

Data Cleaning — drop corrupt cycles, align sampling, handle gaps ↓

Feature Engineering — capacity-fade, cycle count, voltage/temperature curve features ↓

ML Inference — SoH now + RUL forecast per asset ↓

Rule Engine — risk banding: Healthy / Watch / Critical ↓

Agent Orchestration — Monitor → Diagnose (RAG) → Recommend ↓

Prediction + Recommendation persisted ↓

Dashboard Visualisation — fleet ranking + drill-down + reco panel ↓

User Action — acknowledge / schedule / dismiss ↓

Feedback Loop — actions logged to refine models & prompts

7. AI & ML Approach

AI is applied only where it earns its place. Prediction is done by traditional ML (deterministic and provable); reasoning and language are done by the LLM/RAG layer; safety-critical banding stays with a rule engine to avoid hallucination.

AI type	Where used	Why it is needed
Traditional ML	SoH estimation, RUL prediction	The provable core metric judges measure; fast and credible
Predictive analytics	Degradation-trajectory forecasting	Produces the RUL 'knee' curve — the visual hook

AI type	Where used	Why it is needed
Rule engine	Risk banding, safety thresholds	Deterministic, auditable, no hallucination on safety
RAG	Grounding recommendations in documents	Trust + citations + auditability
LLM	Diagnosis narrative, recommendation, report	Turns numbers into decisions — Business Impact & UX
Agentic AI	Monitor → Diagnose → Recommend	Modular, demonstrable, clean demo narrative
Knowledge graph	Not used	Adds ontology effort for zero marginal demo value at this scope

8. Multi-Agent Design

Three agents only — minimal and reliable. Each has a single responsibility and passes structured output to the next.

Agent	Inputs	Outputs	Passes to
1 · Fleet Health Monitor	Telemetry + ML predictions + rule bands	Ranked list of flagged assets	→ Diagnosis Agent
2 · Diagnosis / RCA	Flagged asset + RAG retrieval	Probable cause, severity, citations	→ Recommendation Agent
3 · Recommendation	Diagnosis + RUL + context	Cited action + exportable report	→ Dashboard

9. Data Strategy

The machine-learning core is trained and validated exclusively on real, public NASA battery-degradation cells — no synthetic data enters the model. Only the fleet 'wrapper' (asset IDs, routes, duty cycles) is illustrative and clearly labelled; it is presentation framing and never feeds a prediction.

Dataset	Purpose	Status	Difficulty
NASA PCoE Li-ion Aging (34 real cells)	Train / validate SoH + RUL	Downloaded & in use	Easy
→ core cells B0005 / B0006 / B0007 / B0018	P1 headline metric	In use	Easy
Oxford Battery Degradation	Cross-dataset validation	Optional / planned	Easy
MIT–Stanford (TRI) cycle-life	Larger RUL benchmark	Optional / planned	Easy–Med
OEM manuals / safety standards	RAG grounding corpus (P3)	Public PDFs — planned	Medium
Fleet metadata (IDs, routes)	Illustrative framing — not in model	Synthesised, labelled	Easy

Primary dataset: *NASA Prognostics Center of Excellence Li-ion Battery Aging Dataset — 34 real 18650 cells cycled to end-of-life; the four room-temperature cells (B0005, B0006, B0007, B0018) form the standard SoH/RUL benchmark used for our headline result.*

Integrity note for judges: *all predictions run on real cell data (defensible error metrics); only the fleet wrapper is illustrative, and we state this openly.*

10. Technology Stack

Layer	Choice	Why
Frontend	React + Vite + Tailwind + Recharts	Fast scaffold, strong charts
Backend	FastAPI (Python) + Uvicorn	Same language as ML; rapid REST
Database	SQLite (Postgres if hosted)	Zero-config, file-based
Vector DB	ChromaDB	Local, free, trivial retrieval
ML	scikit-learn + XGBoost	Reliable, fast, provable
LLM	Groq free tier / Ollama (local)	Zero cost; no GPU needed
Embeddings	sentence-transformers (MiniLM)	Local, free
Visualisation	Recharts / Plotly	Degradation curves, gauges
Deployment	Localhost + ngrok / HF Spaces	Zero cost, demo-reliable

Deliberately avoided: paid APIs, streaming infrastructure, Kubernetes, real auth providers, graph databases — none add judging value at this scope.

11. Development Roadmap

Phased for a multi-day build; the good-to-haves are gated behind a green core.

Phase	Objective	Output	Time
P0 Setup	Repo, envs, data download	Running skeleton	0.5 day
P1 ML Core	Provable SoH + RUL	Model + RMSE metric	1.5 days
P2 Backend + DB	Serve predictions	Live prediction API	1 day
P3 RAG + Agents	Grounded recommendation	Cited output	1.5 days
P4 Frontend	Decision surface	Working dashboard	1.5 days
P5 Integration	End-to-end wiring	Full flow works	0.5 day
P6 Demo Polish	Script, rehearse, deck, video	Deliverables ready	1 day

Critical path: P1 → P2 → P3 → P5 → P6. The frontend (P4) is built against mock data in parallel so it is ready when real endpoints land.

12. Risk Analysis & Mitigation

Risk	Type	Mitigation
Weak model accuracy	Model	XGBoost baseline day 1; validate on Oxford; report gain vs. naive baseline
LLM hallucinates unsafe advice	Model	Rule engine owns safety bands; LLM only phrases; show citations
Thin RAG corpus	Data	Pre-assemble 20–40 PDFs + synthesised SOPs before P3

Risk	Type	Mitigation
FE/BE integration crunch	Integration	Freeze API contract in P2; frontend on mock data early
LLM latency in demo	Technical	Free fast inference + cache demo-asset responses
Budget / API cost	Resource	Every dependency is free — no paid service anywhere
Scope creep	Timeline	Hard cuts fixed in Section 4; good-to-haves gated

13. Expected Outcomes & Success Metrics

Aligned to the Problem Statement's evaluation focus, success is measured — not asserted:

- **Prediction accuracy —** SoH / RUL prediction error (RMSE) versus observed degradation, benchmarked against a naive persistence baseline.
- **Warning lead time —** Lead time (in cycles) between a Critical flag and the actual end-of-life threshold.
- **Grounding rate —** Share of recommendations with a valid document citation (target: 100%).
- **Business impact —** Illustrative impact model (assumptions explicit) quantifying avoided premature replacement and downtime.

Achieved in Phase P1 (real NASA data, cross-cell test on unseen cell B0018): SoH estimation RMSE 0.0275 (R^2 0.89); RUL RMSE 16.5 cycles (R^2 0.74). The model never saw the test cell during training — these are genuine generalisation results on 100% real data.

Illustrative impact model (assumptions stated openly):

Assumption	Value
Fleet size (industrial EVs)	100 vehicles
Battery pack cost per vehicle	₹8–12 lakh
Premature replacements avoided (APM)	15–20%
Unplanned battery downtime reduced	~25%
Indicative annual saving (illustrative)	₹1.2–1.8 crore / fleet

Figures are illustrative and clearly assumption-based, as expected of a hackathon impact model — the method, not the exact number, is the deliverable.

14. Alignment with Judging Criteria

Criterion (weight)	Focus	How CellSense AI addresses it
Innovation (25%)	High	Decision-Intelligence loop — grounded, cited APM actions, not just a prediction
Business Impact (25%)	High	Avoided replacement + downtime, quantified; net-zero relevance
Technical Excellence (20%)	High	Real-data ML with measured RMSE; clean ML + rule + RAG separation

Criterion (weight)	Focus	How CellSense AI addresses it
Scalability (15%)	Medium	Architecture generalises to any fleet; stateless services
User Experience (15%)	High	Operator-first dashboard: risk ranking, drill-down, cited action

15. Team & Roles

Team of up to four; roles map one-to-one to the folder structure to avoid collisions. (Replace with your members.)

Member	Role	Responsibility
[Name 1]	ML Lead	Data pipeline, SoH/RUL models, evaluation (critical path)
[Name 2]	Backend + Agents	FastAPI, database, agent orchestration, RAG wiring
[Name 3]	Frontend	React dashboard, charts, drill-down, recommendation panel
[Name 4]	AI/NLP + Integration + Demo	RAG corpus, LLM prompting, report generation, demo & deck

16. Conclusion

CellSense AI is scoped to win: a provable machine-learning core that earns trust, wrapped in a natural-language decision-intelligence layer that delivers genuine Asset Performance Management value. It is built entirely on free, public data with an open-source stack, is realistic for a small team within the hackathon timeline, and maps directly to the highest-weighted judging criteria. Prediction earns credibility; the grounded, cited decision earns the win.

Appendix — Primary data sources: *NASA Prognostics Center of Excellence (battery aging); University of Oxford Battery Degradation Dataset; MIT–Stanford (Toyota Research) cycle-life dataset; public OEM manuals and lithium-ion safety standards.*